COMPUTER NETWORKS (MIC23) — UNIT V

Transport Layer & Application Layer — Handwritten Study Notes

Student Blue-Pen Notebook Style • Short & Direct 2M Answers • Complete Visual Diagrams for Every Question • All Source Papers

SECTION A: 2-MARK QUESTIONS & ANSWERS

Short, direct answers (~40–65 words) written in simple English with student pen-style diagrams

Q1. *(Paper-1 & Mid-2)*

Differentiate between connection-oriented and connectionless transport protocols?

Definition:

- **Connection-Oriented:** The sender and receiver first talk to set up a path before transmitting data (like a phone call).
- **Connectionless:** The sender sends packets immediately without setting up any prior path (like a postcard).

Feature	Connection-Oriented (TCP)	Connectionless (UDP)
Setup Phase	Requires 3-way handshake to connect.	Zero setup; transmits immediately.
Reliability	100% reliable; uses acknowledgements.	Unreliable; does not guarantee delivery.
Speed	Slower due to extra check bits.	Very fast with minimum delay.

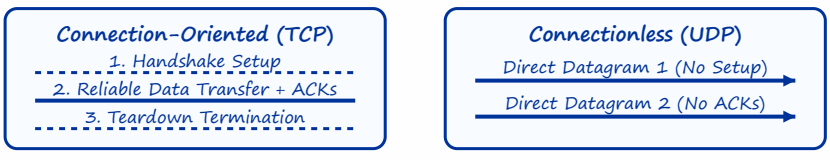


Figure: Connection-Oriented (Setup & ACK) vs Connectionless (Direct Transmit)

Q2. *(Paper-1 & Mid-2)*

Explain the structure of a UDP segment?

Definition:

A **UDP Segment** consists of a small, fixed **8-byte Header** followed by the actual user data payload.

The 4 Header Fields (2 bytes each = 8 bytes total):

- **Source Port (16 bits):** Port number of the sending software program.
- **Destination Port (16 bits):** Port number of the receiving application (e.g., Port 53 for DNS).
- **Length (16 bits):** Total length of UDP header plus payload in bytes.
- **Checksum (16 bits):** Optional error-detection field checking if bits were damaged.

Source Port (16 bits / 2 Bytes)	Destination Port (16 bits / 2 Bytes)
Total Length (16 bits / 2 Bytes)	Checksum (16 bits / 2 Bytes)
Data / Payload (Variable Size) [Fixed Header = 8 Bytes Total]	

Figure: UDP Segment Structure (8-Byte Header)

Q3. *(Paper-2)*

What are the key features of TCP?

Definition:

TCP (Transmission Control Protocol) is the main connection-oriented transport protocol that guarantees ordered and reliable delivery of data.

Key Features:

- **Process-to-Process:** Delivers data directly to the correct running program using 16-bit port numbers.
- **Reliable Delivery:** Uses sequence numbers, acknowledgements (ACK), and retransmissions so no packet is lost.
- **Flow & Congestion Control:** Matches transmission speed to receiver buffer capacity and protects network routers.

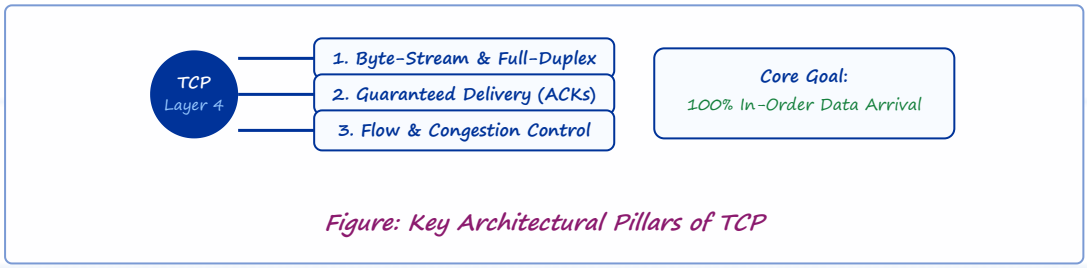


Figure: Key Architectural Pillars of TCP

Q4. *(Paper-2)*

How does TCP achieve error control?

Definition:

Error Control *in TCP ensures that all data arrives at the receiver without bit corruption, without duplicates, and with zero missing bytes.*

Three Methods Used:

- **Checksum:** *A 16-bit checksum detects damaged bits; bad segments are discarded silently.*
- **Cumulative ACK:** *The receiver tells the sender the next byte number it expects to receive.*
- **Retransmissions:** *If the retransmission timer (RTO) expires or 3 duplicate ACKs arrive, TCP resends the missing data immediately.*

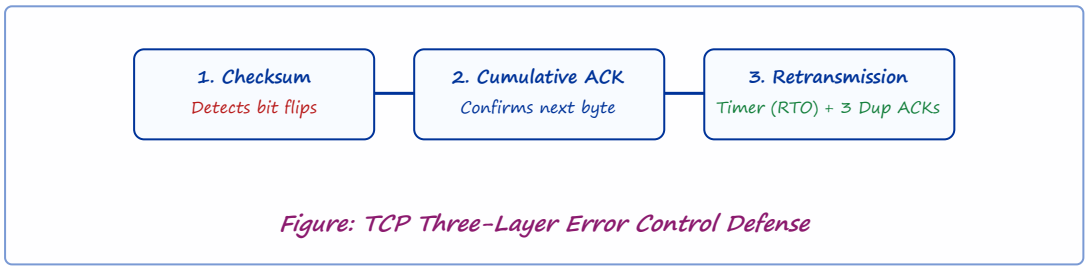


Figure: TCP Three-Layer Error Control Defense

Q5. *(Paper-3 & Mid-2)*

What is the role of the transport layer in the OSI model?

Definition:

The **Transport Layer (Layer 4)** provides true end-to-end, **process-to-process delivery** between specific software programs running on two computers.

Main Roles:

- **Port Addressing:** Attaches port numbers (e.g., 80 for HTTP, 53 for DNS) so data reaches the exact software application.
- **Segmentation & Reassembly:** Splits large messages into smaller segments at sender and puts them back in order at receiver.
- **Reliability & Flow Control:** Prevents data loss and manages buffer space.

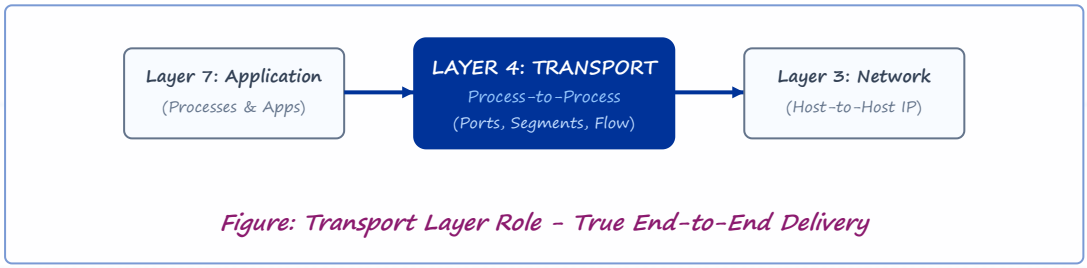


Figure: Transport Layer Role - True End-to-End Delivery

Q6. *(Paper-3 & Mid-2)*

Differentiate between local and remote logging.

Definition:

- **Local Logging:** A user logs in directly using a keyboard and screen physically connected to the computer.
- **Remote Logging:** A user sits at a local PC and logs into a server located far away across a network using programs like TELNET or SSH.

Feature	Local Logging	Remote Logging
Location	User is physically at the machine.	User is across a network (miles away).
Network	No network connection needed.	Requires LAN or Internet connection.
Software	Handled directly by local OS.	Uses client-server tools (TELNET, SSH).

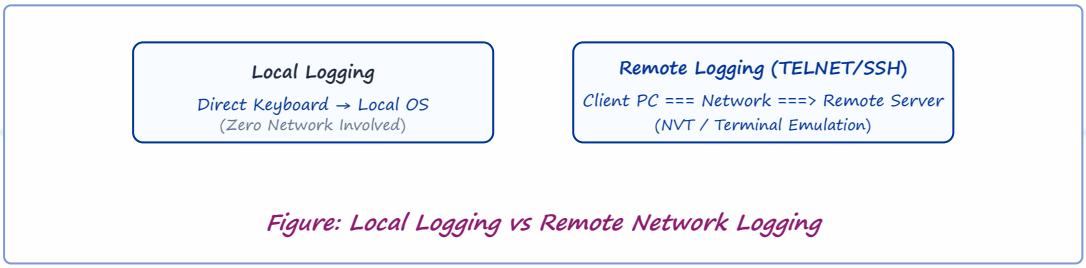


Figure: Local Logging vs Remote Network Logging

Q7. *(Mid-2)*

Name two services provided by the UDP protocol.

Definition:

UDP (User Datagram Protocol) is a lightweight, connectionless transport protocol designed for maximum speed.

Two Services:

- **Process-to-Process Delivery (Port Addressing):** Uses 16-bit port numbers (e.g., Port 53 for DNS, Port 67 for DHCP) so data reaches the correct running program.
- **Error Detection (Checksum):** Includes an optional 16-bit checksum to detect damaged bits and drop corrupted packets.

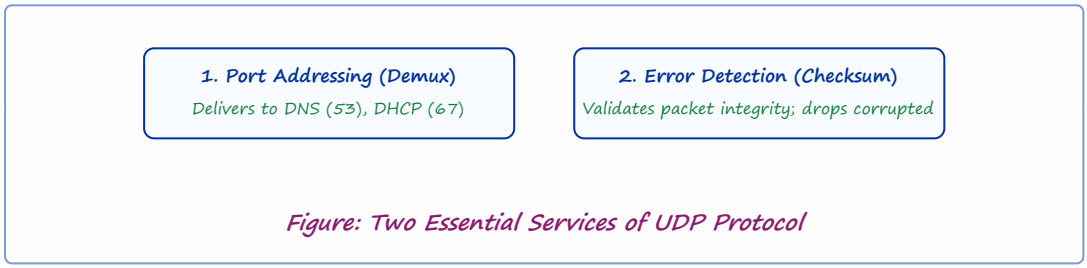


Figure: Two Essential Services of UDP Protocol

Q8. *(Mid-2)*

Define TELNET in the context of the Application layer.

Definition:

TELNET (Teletype Network) is an Application Layer client-server protocol that lets a user log into and control a remote computer across a network as if sitting directly in front of it.

Key Points:

- **Port Number:** Operates over TCP well-known port **23**.
- **NVT Concept:** Uses **Network Virtual Terminal (NVT)** character formatting so different operating systems can talk.
- **Limitation:** Insecure because passwords travel as plain text (replaced by secure SSH today).

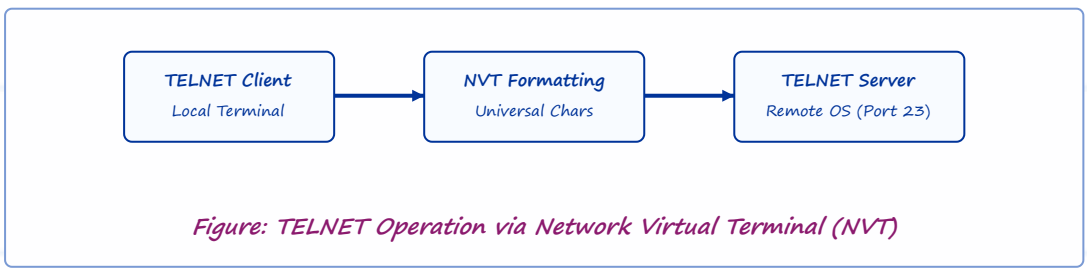


Figure: TELNET Operation via Network Virtual Terminal (NVT)

SECTION B: 5-MARK QUESTIONS (FULL 10-MARK EXAM STRUCTURE)

Detailed answers with Introduction, Dual Diagrams (Standard Exam + Intuitive Analogy), Box Comparison Tables, and Real-Life Examples

Q1. *(Paper-1)*

Describe client-server communication and HTTP request message format?

1. Introduction

Modern Internet applications work on the **Client-Server Architecture**. In this model, one computer (the client) asks for services, and another computer (the server) provides them. The **Hypertext Transfer Protocol (HTTP)** is the foundation protocol used on the World Wide Web to exchange web pages between clients and servers.

2. Client-Server Communication Concept

- **Client (Web Browser):** Runs on user devices (Chrome, Firefox, Edge). It creates an HTTP request when a user types a URL or clicks a link.
- **Server (Web Server):** A powerful computer (Apache, Nginx) running 24/7 that stores website files (HTML, CSS, images) and listens for requests.
- **Request-Response Cycle:**
  - Client opens a TCP connection to the server (default port 80 or 443).
  - Client sends an **HTTP Request Message**.
  - Server processes the request and sends back an **HTTP Response Message**.
  - Client renders the web page on screen.

3. Detailed HTTP Request Message Format

An HTTP Request message consists of 4 parts:

- **Request Line:** Contains the HTTP Method, URL path, and HTTP version.
- Example: `GET /index.html HTTP/1.1`
- **Request Headers:** Key-value pairs providing extra details about the client.
- `Host: www.example.com` (Target server domain).
- `User-Agent: Mozilla/5.0` (Client browser identity).
- `Accept: text/html` (File formats the client can read).
- **Empty Line (CRLF):** A mandatory blank line separating headers from the body.

• **Message Body (Entity Body):** Optional data sent to the server (used with POST or PUT methods to submit forms or upload files).

4. Common HTTP Request Methods

- **GET:** Retrieves data/webpages from the server (does not change server state).
- **POST:** Sends new data to the server (e.g., submitting login details).
- **PUT:** Uploads or replaces an existing resource on the server.
- **DELETE:** Removes a specified file or resource from the server.

5. Visual Examination & Concept Diagrams

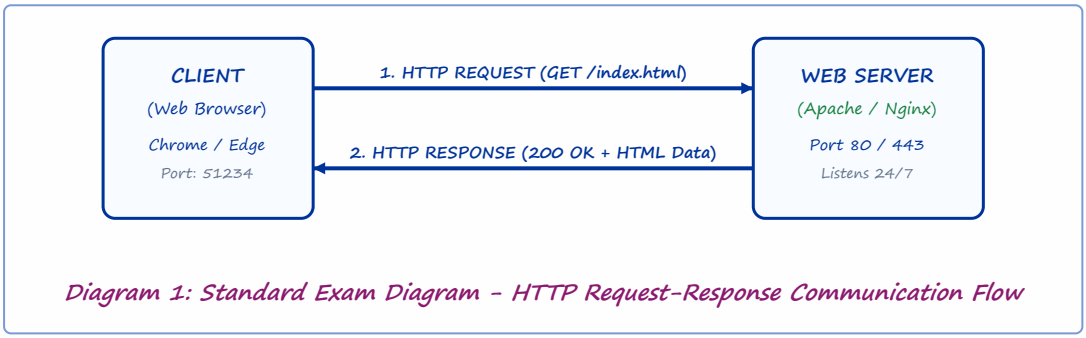


Diagram 1: Standard Exam Diagram - HTTP Request-Response Communication Flow

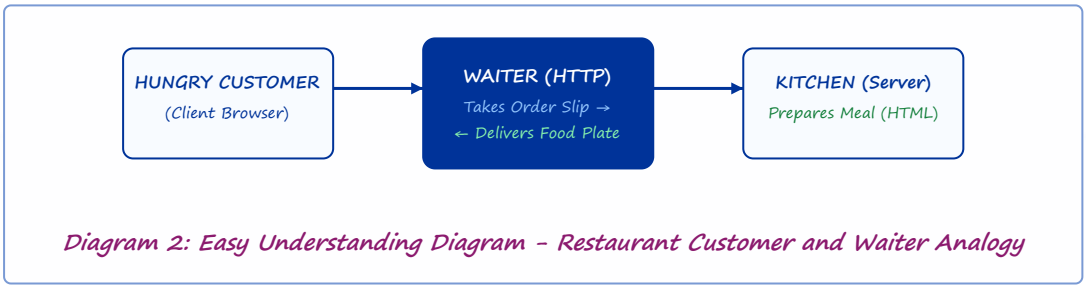


Diagram 2: Easy Understanding Diagram - Restaurant Customer and Waiter Analogy

6. Real-Life Example

When you type `www.amazon.com/shoes` in your browser:

- Your browser sends `GET /shoes HTTP/1.1` with host `www.amazon.com`.
- The Amazon web server locates the shoe catalog in its database and sends back HTML code with pictures.

7. Comparison of HTTP Methods

HTTP Method	Primary Purpose	Sends Data In Body?	Safe/Idempotent?
GET	Retrieve an existing webpage or file.	No (parameters in URL).	Yes (does not alter server).

POST	Submit sensitive form data or files.	Yes (in request body).	No (creates new records).
PUT	Update or completely replace data.	Yes.	Yes (repeated calls give same result).
DELETE	Remove data or file from server.	Usually No.	Yes.

8. Conclusion

The client-server model combined with HTTP request-response messaging provides a clean, standardized, and scalable way for billions of devices to browse web pages across the world.

Q2. *(Paper-1)*

Explain different services provided by Transport layer?

1. Introduction

The **Transport Layer (Layer 4)** is the central core of the computer networking stack. While the Network Layer only moves packets between computers (host-to-host), the Transport Layer delivers data to the **exact application program** running inside the computer (process-to-process).

2. Key Services Provided by Transport Layer

1. Port Addressing (Service Point Addressing)

- A computer runs dozens of programs at the same time (browser, email, Spotify, game).
- The Transport layer attaches a **16-bit Port Number** (ranging from 0 to 65535) to every packet so the operating system delivers incoming data to the exact program.

2. Segmentation and Reassembly

- Application layer messages can be huge (e.g., a 100 MB video).
- Transport layer splits the message into manageable chunks called **segments**, labels them with sequence numbers, and reassembles them in original order at the receiver.

3. Connection Control

- **Connection-Oriented Service (TCP):** Establishes a 3-way handshake connection, transfers data reliably, and cleanly tears down the connection.

- **Connectionless Service (UDP):** Transmits packets independently without setting up a connection, saving time.

4. Flow Control

- Prevents a fast sender from drowning a slow receiver with too much data.
- Uses the **Sliding Window Protocol** where the receiver advertises how much buffer space it has left (`Window Size`).

5. Error Control

- Ensures data arrives without bit errors, lost packets, or duplicates.
- Uses **Checksums**, **Acknowledgements (ACK)**, and **Retransmission Timers (RTO)**.

6. Congestion Control

- Prevents too many packets from clogging network routers and switches across the global Internet.

3. Visual Diagrams

Diagram 1: Standard Exam Diagram (Process-to-Process Port Multiplexing)

Diagram 2: Easy Understanding Diagram (Apartment Building Mail Delivery Analogy)

4. Summary Table of Transport Layer Services

Service	Why It Is Needed	How It Is Implemented
Port Addressing	Distinguish between multiple running apps.	16-bit Port numbers (0 to 65,535).
Segmentation	Allow big files to travel over network limits.	Sequence numbering and reassembly buffers.
Flow Control	Protect slow receivers from buffer overflow.	Sliding window method.
Error Control	Ensure zero lost or corrupted bytes.	Checksum, ACK, and timeout retransmissions.
Congestion Control	Protect Internet routers from traffic collapse.	Slow-start and congestion avoidance algorithms.

5. Conclusion

Without the Transport Layer, computers could send data between network cards, but the operating system would not know which app should open the data, leading to total communication breakdown.

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5. Visual Examination & Concept Diagrams

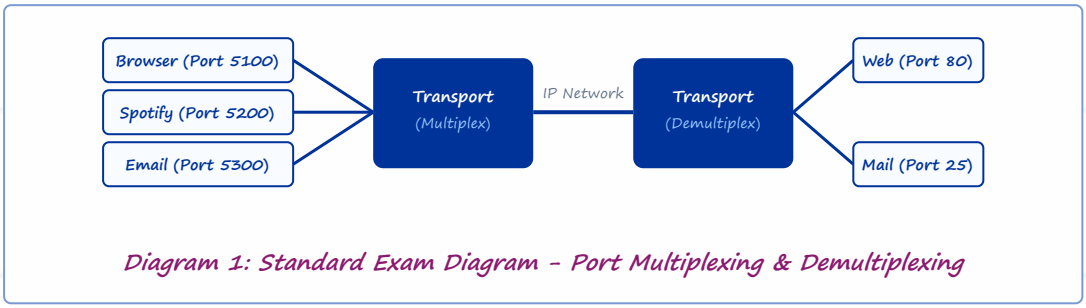


Diagram 1: Standard Exam Diagram - Port Multiplexing & Demultiplexing

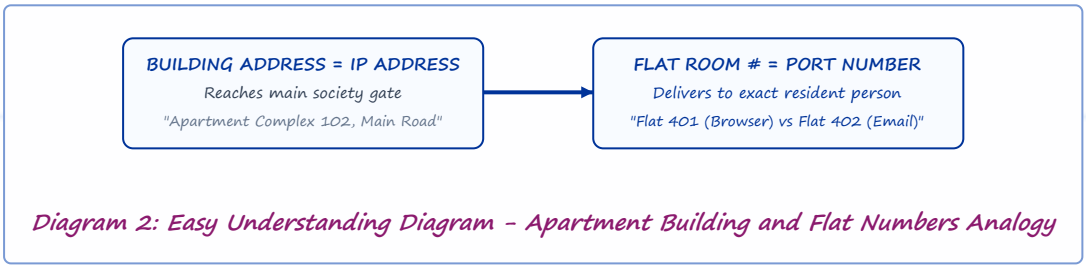


Diagram 2: Easy Understanding Diagram - Apartment Building and Flat Numbers Analogy

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Q3. *(Paper-1)*

How does TCP ensure reliable data transmission? Explain.

1. Introduction

The underlying IP network is "unreliable" (best-effort)—it can drop packets, duplicate them, delay them, or deliver them out of order. **TCP (Transmission Control Protocol)** provides 100% reliable data transmission on top of this unreliable network using four core methods.

2. The Four Pillars of TCP Reliability

1. Sequence Numbers & Byte-Stream Tracking

- TCP views data as an continuous stream of bytes. Every single byte is assigned a unique 32-bit sequence number.
- If segments arrive out of order, the receiver uses sequence numbers to put them back into perfect order before giving them to the application.

2. Positive Acknowledgements (ACK) & Cumulative ACKs

- Whenever the receiver gets valid data, it sends back an **Acknowledgement Number**.
- TCP uses **cumulative ACKs**: An ACK number of `N` means: *"I have safely received all bytes up to N-1, and I am now waiting for byte N."*

3. Retransmission Mechanisms

- **Retransmission Timeout (RTO):** When TCP sends a segment, it starts a timer. If the timer expires before an ACK arrives, the segment is automatically resent.
- **Fast Retransmit:** If 1 segment is missing while later ones arrive, the receiver keeps sending duplicate ACKs for the missing byte. When the sender gets **3 duplicate ACKs**, it retransmits the missing segment immediately without waiting for the timer.

4. Checksum for Error Detection

- Every TCP segment has a 16-bit Checksum.
- If noise flips a bit in transit, the receiver detects it and silently throws away the bad segment. The sender will time out and retransmit a fresh copy.

3. Connection Life Cycle (Establishment and Termination)

- **3-Way Handshake Connection:** `SYN` o `SYN + ACK` o `ACK`. Establishes starting sequence numbers and buffer sizes.
- **4-Way Handshake Termination:** `FIN` o `ACK` o `FIN` o `ACK`. Safely closes data channels in both directions.

4. Visual Diagrams

Diagram 1: Standard Exam Diagram (TCP 3-Way Handshake & Sequence ACK Flow)

Diagram 2: Easy Understanding Diagram (Courier Tracking & Receipt Analogy)

5. Real-Life Example

Imagine sending a 3-page letter where pages arrive as Page 1, Page 3, Page 2.

- UDP would read it jumbled.
- TCP reads the page numbers, rearranges them to Page 1, 2, 3, and if Page 2 blew away in the wind, TCP asks the sender to reprint and mail Page 2 again.

5. Visual Examination & Concept Diagrams

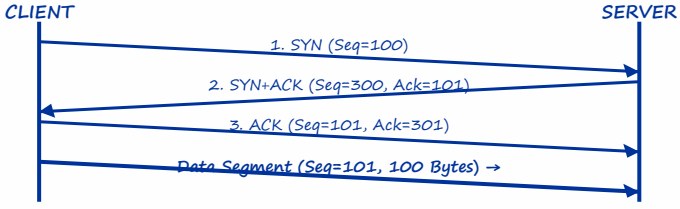


Diagram 1: Standard Exam Diagram - TCP 3-Way Handshake and Data Transfer Flow



Diagram 2: Easy Understanding Diagram – Registered Post with Delivery Signature Analogy

6. Summary Table

Mechanism	Problem It Solves	Action Taken
Sequence Numbers	Out-of-order arrival and duplicates.	Reorders packets; discards duplicate copies.
16-bit Checksum	Bit corruption during cable transit.	Discards damaged segment immediately.
RTO Timer	Lost packets in congested routers.	Automatically retransmits when timer expires.
3 Duplicate ACKs	Long waiting time for timer.	Triggers Fast Retransmit immediately.

7. Conclusion

Through byte sequence numbers, positive acknowledgements, checksums, and intelligent retransmissions, TCP guarantees that what the sender transmitted is 100% identical to what the receiver receives.

Q4. *(Paper-1)*

Explain Domain Name System (DNS)?

1. Introduction

Computers across the Internet locate each other using numeric **IP Addresses** (e.g., `142.250.190.46`). However, human beings cannot remember long strings of numbers; we prefer readable names like `google.com`. The **Domain Name System (DNS)** is the Internet's telephone directory that translates human-friendly domain names into computer-friendly IP addresses.

2. How DNS Works (Tree-like Name Space (Hierarchical))

DNS is structured as an inverted tree with the **Root** at the top:

- **Root DNS Servers (.):** The top level of the hierarchy (13 root server clusters named A through M).
- **Top-Level Domain (TLD) Servers:** Manage domains ending in `.com`, `.org`, `.net`, `.edu`, or country codes like `.in`, `.uk`.
- **Authoritative DNS Servers:** The final servers that hold the actual official IP mapping for a specific website (e.g., the DNS server for `google.com`).
- **Local DNS Resolver (Recursive Resolver):** Usually operated by your ISP (Internet Service Provider) or Google (`8.8.8.8`). It does all the searching work on behalf of your browser.

3. Step-by-Step DNS Resolution Process

When you type `www.example.com` into your browser:

- **Browser Cache Check:** Browser checks if it already knows the IP. If not, it asks the Local DNS Resolver.
- **Query Root Server:** Local Resolver asks the Root Server: *"Where is .com?"* Root responds with the address of the `.com` TLD Server.
- **Query TLD Server:** Local Resolver asks the `.com` TLD Server: *"Where is example.com?"* TLD responds with the Authoritative Name Server address.
- **Query Authoritative Server:** Local Resolver asks the Authoritative Server: *"What is the IP of www.example.com?"* The server replies: `93.184.216.34`.
- **Return to Browser:** The Local Resolver caches the IP and gives it to your browser. Your browser now connects directly to the web server.

4. Visual Diagrams

Diagram 1: Standard Exam Diagram (DNS Hierarchical Query Resolution Steps)

Diagram 2: Easy Understanding Diagram (Phone Contact Book Analogy)

5. Common DNS Record Types

Record Type	Full Name	Function / Description
A Record	Address Record	Maps a domain name to an **IPv4 Address** (32-bit).
AAAA Record	IPv6 Address Record	Maps a domain name to an **IPv6 Address** (128-bit).
CNAME	Canonical Name	Maps an alias/nickname to the real name (`www.site.com` o `site.com`).

MX Record	Mail Exchange	Points to the mail server responsible for receiving emails for the domain.

5. Visual Examination & Concept Diagrams

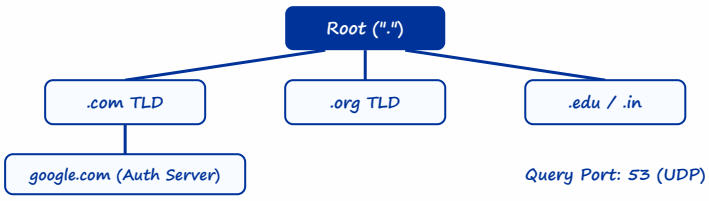


Diagram 1: Standard Exam Diagram – Hierarchical DNS Inverted Tree Architecture



Diagram 2: Easy Understanding Diagram – Smartphone Contacts Directory Analogy

6. Why DNS Uses UDP Port 53

DNS queries usually consist of a single request packet and a single response packet. Setting up a 3-way TCP handshake would double the lookup time. Therefore, standard DNS queries use **UDP on Port 53** for blazing-fast lookups. (TCP is only used for zone transfers or responses larger than 512 bytes).

7. Conclusion

DNS is a very important service of the Application Layer. It allows billions of users to seamlessly navigate the Internet using easily memorable names while computers route traffic using IP addresses.

Q5. *(Paper-2)*

Describe the Structure of TCP segment Header?

1. Introduction

Data transferred by the Transmission Control Protocol is packaged into units called **TCP Segments**. Every segment consists of a **Header** followed by the user data. The TCP header has

a minimum size of **20 bytes** (when no options are present) and can extend up to **60 bytes** with options.

2. Detailed Explanation of Header Fields (32 bits / 4 bytes per row)

• **Source Port Address (16 bits):** Port number of the sending application (e.g., 51234).

• **Destination Port Address (16 bits):** Port number of the receiving application (e.g., Port 80 for Web).

• **Sequence Number (32 bits):** Defines the number assigned to the first data byte in this segment. Used to reconstruct out-of-order data.

• **Acknowledgement Number (32 bits):** Valid if ACK flag is set. Contains the next byte number the receiver is expecting.

• **Header Length (HLEN / Data Offset) (4 bits):** Stores the number of 4-byte words in the header. (Value 5 means 5 imes 4 = 20 ext{ bytes}; maximum is 15 meaning 60 bytes).

• **Reserved (6 bits):** Reserved for future use (always set to 0).

• **Control Flags (6 bits - 1 bit each):**

- **URG (Urgent):** Indicates urgent pointer field is valid.
- **ACK (Acknowledgement):** Confirms acknowledgement number is valid.
- **PSH (Push):** Tells receiver to push data to application immediately.
- **RST (Reset):** Aborts and resets a faulty connection.
- **SYN (Synchronize):** Used during 3-way handshake to synchronize sequence numbers.
- **FIN (Finish):** Used to terminate the connection.

• **Window Size (16 bits):** Advertises the receiver's available buffer space in bytes (implements flow control).

• **Checksum (16 bits):** Mandatory error detection field covering header and payload.

• **Urgent Pointer (16 bits):** Points to the end of urgent data when URG flag is set.

• **Options (0 to 40 bytes):** Used for Maximum Segment Size (MSS), window scaling, and timestamps.

3. Visual Diagrams

Diagram 1: Standard Exam Diagram (TCP 20-Byte Header Format)

Diagram 2: Easy Understanding Diagram (Courier Shipping Label Analogy)

4. Summary Table of Key Control Flags

Flag	Name	When Set to 1

SYN	Synchronize	Sent in the very first packet to start a connection.
ACK	Acknowledge	Confirms receipt of data (set in almost all TCP packets).
FIN	Finish	Sent when an application has no more data to send and wants to close.
RST	Reset	Immediately kills a connection if an error occurs.

5. Conclusion

The TCP segment header contains comprehensive fields for addressing, sequencing, acknowledgement, flow control, and error handling, making it the most dependable protocol on the Internet.

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5. Visual Examination & Concept Diagrams

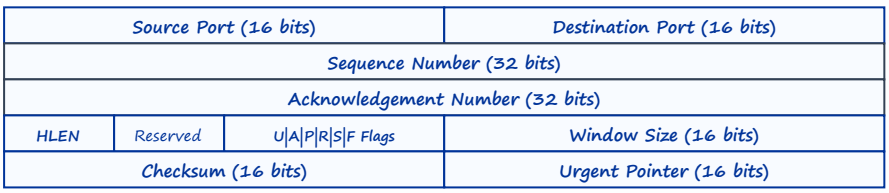


Diagram 1: Standard Exam Diagram – TCP Segment Header (20-Byte Base Format)



Diagram 2: Easy Understanding Diagram – Courier Shipping Label Analogy

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Q6. *(Paper-2)*

Explain the architectural view of WWW?

1. Introduction

The **World Wide Web (WWW)** is a massive, distributed client-server information system that allows users to access linked documents (webpages), multimedia, and applications over the Internet. It was invented in 1989 by Tim Berners-Lee.

2. Three Fundamental Components of WWW

The entire architecture of WWW relies on three core building blocks:

• **Addressing & Identification (URI / URL):** Uniform Resource Locator defines the unique global address of any resource on the web.

- Syntax: `protocol://host:port/path?query` (e.g., `https://www.google.com/search?q=networks`).

• **Standard Document Format (HTML / CSS / JS):**

- **HTML (Hypertext Markup Language):** Provides the basic skeleton and hyperlinks.
- **CSS (Cascading Style Sheets):** Controls visual presentation, styling, and layouts.
- **JavaScript:** Adds interactivity and dynamic behavior.

• **Transfer Protocol (HTTP / HTTPS):**

- The communication language spoken between browser and web server to request and deliver pages.

3. Modern 3-Tier Architecture of WWW

Real-world web systems operate on a **3-Tier Architecture**:

- **Tier 1: Client Tier (Presentation Layer):** The user's web browser that executes HTML/CSS/JavaScript and renders the visual interface.
- **Tier 2: Application Tier (Business Logic Layer):** Web application servers (Node.js, Django, Java Spring, PHP) that process user input, calculate results, and handle security.
- **Tier 3: Database Tier (Data Storage Layer):** Storage engines (MySQL, PostgreSQL, MongoDB) that permanently store customer accounts, products, and passwords.

4. Visual Diagrams

Diagram 1: Standard Exam Diagram (3-Tier Architecture of World Wide Web)

Diagram 2: Easy Understanding Diagram (Restaurant Ordering Analogy)

5. Comparison: Static Web Pages vs Dynamic Web Pages

Feature	Static Web Pages	Dynamic Web Pages
Content	Same content shown to every visitor.	Customized content generated on the fly.
Server Work	Server simply reads an HTML file from disk.	Server executes code and queries database.
Speed	Extremely fast and easy to cache.	Takes slightly more time to compute and assemble.
User Interaction	Read-only; no personalized login.	Full interaction (shopping carts, user profiles).

Example	Personal resume or documentation page.	Amazon, Instagram, YouTube dashboard.

5. Visual Examination & Concept Diagrams



Diagram 1: Standard Exam Diagram – 3-Tier Architecture of World Wide Web



Diagram 2: Easy Understanding Diagram – Restaurant Dining, Chef, and Pantry Analogy

6. Conclusion

The World Wide Web's 3-tier architecture separates visual presentation, computing logic, and permanent storage, allowing modern websites to serve millions of simultaneous users reliably.

Q7. *(Paper-2)*

Describe the architecture of E-Mail and its services?

1. Introduction

Electronic Mail (E-Mail) is one of the oldest and most vital network applications. It allows users to compose, send, store, and receive digital messages across the Internet. Unlike instant messaging, e-mail is a **store-and-forward system**—the recipient does not need to be online when the sender sends the message.

2. Architecture & Components of E-Mail

An e-mail system consists of three main components:

- **User Agent (MUA – Mail User Agent):**
- The client software used by the human to read, write, and reply to messages (e.g., Outlook, Apple Mail, or Webmail like Gmail).

- **Mail Transfer Agent (MTA):**
- The server software running 24/7 that transfers emails between servers using **SMTP** (e.g., Postfix, Sendmail).
- **Mail Delivery Agent (MDA) / Message Access Agent (MAA):**
- Software that accepts incoming mail from the MTA and puts it into the recipient's mailbox. The user then pulls the mail using **POP3** or **IMAP**.

3. Protocols Used in E-Mail

- **SMTP (Simple Mail Transfer Protocol - Port 25/587):** A "Push" protocol used to upload emails from user to server, and forward emails between servers.
- **POP3 (Post Office Protocol version 3 - Port 110):** A "Pull" protocol that downloads emails from server to local PC and typically deletes them from the server.
- **IMAP (Internet Message Access Protocol - Port 143):** An advanced "Pull" protocol that leaves emails on the server and synchronizes folders across multiple devices (phone, laptop).

4. Visual Diagrams

Diagram 1: Standard Exam Diagram (End-to-End E-Mail Architecture & Protocols)

Diagram 2: Easy Understanding Diagram (Post Office Mail Van Analogy)

5. Comparison: POP3 vs IMAP

Feature	POP3 (Post Office Protocol)	IMAP (Internet Message Access)
Storage Location	Downloads emails to local PC; deletes from server.	Keeps all emails on the server permanently.
Multiple Devices	Difficult; read status does not sync.	Perfect sync across phone, laptop, and web.
Offline Access	Full access to downloaded messages.	Partial access (must cache messages locally).
Server Disk Space	Consumes very little server disk space.	Requires large server storage.
Port Numbers	Port 110 (Insecure), Port 995 (SSL).	Port 143 (Insecure), Port 993 (SSL).

5. Visual Examination & Concept Diagrams

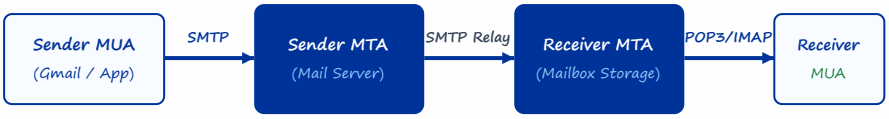


Diagram 1: Standard Exam Diagram – End-to-End E-Mail Architecture (SMTP Push & POP3/IMAP Pull)



Diagram 2: Easy Understanding Diagram – Postal Mail Van and Letter Box Analogy

6. Conclusion

E-mail relies on a smart combination of push protocols (SMTP) to move messages across the globe and pull protocols (POP3/IMAP) to let users retrieve messages on demand.

Q8. *(Paper-2)*

Describe HTTP Response message format?

1. Introduction

When an HTTP client (web browser) sends an HTTP Request to a web server, the server processes the request and replies with an **HTTP Response Message**. The response contains the status of the operation (success, error, redirect) along with the requested HTML page, image, or file.

2. Structure of HTTP Response Message

An HTTP Response message consists of 4 distinct parts:

- **Status Line:** The very first line that contains:
- **HTTP Version:** e.g., `HTTP/1.1` or `HTTP/2`.
- **Status Code:** A 3-digit numeric code indicating the outcome (e.g., `200`).
- **Reason Phrase:** A human-readable text explanation (e.g., `OK`).
- *Example:* `HTTP/1.1 200 OK`
- **Response Headers:** Metadata providing details about the server and data:

- `Content-Type: text/html` (File format of the payload).
- `Content-Length: 1024` (Size of the payload in bytes).
- `Server: Apache/2.4.41` (Software running on the web server).
- `Date: Wed, 04 Nov 2026 12:00:00 GMT`

• **Empty Line (CRLF):** A required blank line that marks the end of headers.

• **Message Body (Entity Body):** The actual payload requested by the client (the HTML code, image binary, or JSON data).

3. Common HTTP Status Code Categories

Code Range	Category Meaning	Common Examples
1xx	Informational	`100 Continue` (Server received request headers).
2xx	Success	`200 OK` (Request succeeded, data included).
3xx	Redirection	`301 Moved Permanently` (Page relocated to new URL).
4xx	Client Error	`400 Bad Request`, `404 Not Found` (File does not exist).
5xx	Server Error	`500 Internal Server Error`, `503 Service Unavailable`.

4. Visual Diagrams

Diagram 1: Standard Exam Diagram (HTTP Response Format Breakdown)

Diagram 2: Easy Understanding Diagram (Official Reply Letter Analogy)

5. Real-Life Example

- When you open `youtube.com` and it loads normally, the server sends `HTTP/1.1 200 OK`.
- If you mistype the video link as `youtube.com/xyz123fake`, the server replies with `HTTP/1.1 404 Not Found` and a body saying *"This video does not exist."*

5. Visual Examination & Concept Diagrams



Diagram 1: Standard Exam Diagram – HTTP Response Message Structure



Diagram 2: Easy Understanding Diagram - Official University Admission Letter Analogy

6. Conclusion

The HTTP response message structure provides an organized way for servers to communicate success or failure codes along with the actual requested content back to browsers.

Q9. *(Paper-3)*

List five differences between UDP and TCP.

1. Introduction

The Transport Layer provides two primary protocols to applications: **Transmission Control Protocol (TCP)** and **User Datagram Protocol (UDP)**. TCP prioritizes reliability and accuracy, while UDP prioritizes low latency and raw speed.

2. Detailed Box Comparison Table

Feature / Parameter	TCP (Transmission Control Protocol)	UDP (User Datagram Protocol)
1. Connection Type	**Connection-Oriented:** Establishes a 3-way handshake before sending data.	**Connectionless:** Transmits packets immediately without any setup.
2. Reliability & Delivery	**Guaranteed:** Guarantees data arrives safely and in correct order using ACKs.	**Unreliable (Best Effort):** Packets may be lost, duplicated, or out of order.
3. Speed & Performance	**Slower:** High extra header size due to acknowledgements and flow control timers.	**Very Fast:** Low extra header size with zero waiting for acknowledgements.
4. Header Size	**Larger:** Minimum 20 bytes (can grow up to 60 bytes with options).	**Tiny:** Fixed 8-byte header only.
5. Flow & Congestion Control	**Yes:** Built-in sliding window and congestion avoidance algorithms.	**No:** No flow or congestion control; transmits at whatever rate given.

6. Retransmission of Lost Data	**Yes:** Lost packets are automatically detected and retransmitted.	**No:** Lost packets are simply dropped and never resent.
7. Real-Life Examples	Web browsing (HTTP/HTTPS), File transfer (FTP), Email (SMTP).	Live video streaming (YouTube Live), Online gaming, DNS, VoIP calls.

3. Visual Diagrams

Diagram 1: Standard Exam Diagram (Side-by-Side Header & Connection Comparison)

Diagram 2: Easy Understanding Diagram (Phone Call vs Radio Broadcast Analogy)

4. Summary & Recommendation

- Use **TCP** when every single bit matters (money transfers, documents, passwords).
- Use **UDP** when speed and real-time delivery matter more than a few dropped frames (live video calls, multiplayer gaming).

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5. Visual Examination & Concept Diagrams

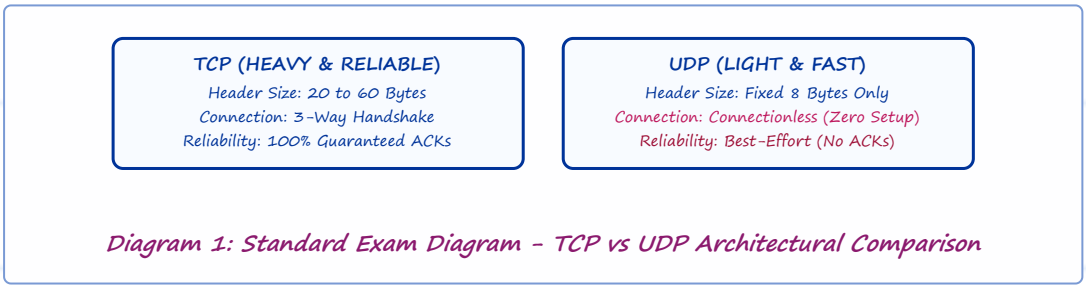


Diagram 1: Standard Exam Diagram - TCP vs UDP Architectural Comparison

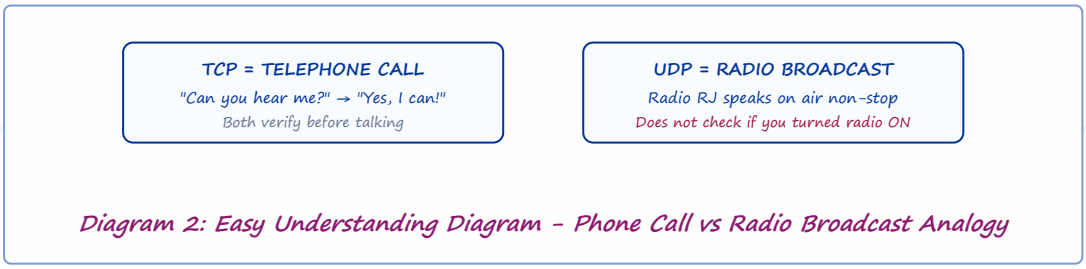


Diagram 2: Easy Understanding Diagram - Phone Call vs Radio Broadcast Analogy

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Q10. *(Paper-3)*

Explain the Transmission Control Protocol (TCP) in detail.

1. Introduction

TCP (Transmission Control Protocol) is the primary transport protocol of the Internet suite (TCP/IP). It operates at Layer 4 and transforms the unreliable, connectionless packet delivery of the IP layer into a **reliable, connection-oriented, ordered byte-stream service**.

2. Key Architectural Characteristics of TCP

- **Connection-Oriented:** A logical connection is negotiated between sender and receiver before any user data flows.
- **Byte-Stream Delivery:** Data is treated as an continuous stream of bytes, packaged into segments governed by the Maximum Segment Size (MSS).
- **Full-Duplex:** Bidirectional data flow occurs at the exact same time across two independent streams.
- **End-to-End Reliability:** Tracks individual bytes using 32-bit sequence numbers and positive acknowledgements.
- **Flow Control:** Implements a dynamic sliding window to protect receiver buffers.
- **Congestion Control:** Senses network bottlenecks and throttles transmission speed.

3. Complete Life Cycle of a TCP Connection

A complete TCP communication goes through three distinct phases:

Phase 1: Connection Establishment (3-Way Handshake)

- **Step 1 (SYN):** Client sends a segment with `SYN=1` and an Initial Sequence Number (`Seq=x`).
- **Step 2 (SYN + ACK):** Server responds with `SYN=1`, `ACK=1`, `Seq=y`, and `Ack=x+1`.
- **Step 3 (ACK):** Client replies with `ACK=1` and `Ack=y+1`. The connection is now open.

Phase 2: Data Transfer Phase

- Both sides exchange segments containing data bytes.
- The sender slides its window forward as cumulative ACKs arrive.
- If a segment's timer expires without an ACK, it is retransmitted.

Phase 3: Connection Termination (4-Way Handshake)

- Client sends `FIN=1` when its application finishes sending.
- Server responds with `ACK` (Client-to-Server channel is closed).
- Server finishes sending its own remaining data and sends `FIN=1`.
- Client responds with `ACK` (Server-to-Client channel is closed).

4. Visual Diagrams

Diagram 1: Standard Exam Diagram (Complete TCP Life Cycle: Handshake, Data & Teardown)

Diagram 2: Easy Understanding Diagram (Business Meeting Phone Call Analogy)

5. Advantages & Disadvantages of TCP

Advantages	Disadvantages
100% accurate data delivery (zero missing bytes).	Higher extra header size (20 to 60-byte header).
Automatic error recovery via retransmissions.	Slower than UDP due to connection setup and ACKs.
Automatic flow and network congestion control.	Cannot support broadcasting or multicasting.

5. Visual Examination & Concept Diagrams



Diagram 1: Standard Exam Diagram – The Three Complete Phases of TCP Connection



Diagram 2: Easy Understanding Diagram – Formal Phone Meeting Analogy

6. Conclusion

TCP is the workhorse of the modern Internet. It provides unmatched reliability, making it the mandatory choice for websites, banking, email, and file transfers.

Q11. *(Paper-3)*

Explain HTTP protocol in detail.

1. Introduction

HTTP (Hypertext Transfer Protocol) is an Application Layer protocol that powers the World Wide Web. It governs how web browsers request information and how web servers deliver web

pages, multimedia, and files. HTTP is fundamentally a **stateless, request-response protocol** that runs on top of TCP (port 80 for HTTP, port 443 for HTTPS).

2. Key Characteristics of HTTP

• **Client-Server Architecture:** Clients (browsers) request resources; servers store and serve resources.

• **Stateless Nature:** The server does not remember anything about previous requests. Each request is treated as brand new. (Websites use **Cookies** and **Sessions** to remember logins and shopping carts).

• **Media Independent:** Any data type can be sent (HTML, plain text, PNG, MP4, JSON) as long as it is specified in the `Content-Type` header (MIME type).

• **Connection Handling:**

- **Non-Persistent HTTP (HTTP/1.0):** Opens a separate TCP connection for every single file. (Very slow).
- **Persistent HTTP (HTTP/1.1):** Keeps a single TCP connection open to download multiple images, CSS, and scripts.

3. Detailed Request and Response Messages

- **HTTP Request:**
- Request Line: `GET /index.html HTTP/1.1`
- Headers: `Host: www.site.com`, `User-Agent: Chrome`
- Body: Optional user data (form input).
- **HTTP Response:**
- Status Line: `HTTP/1.1 200 OK`
- Headers: `Content-Type: text/html`, `Content-Length: 500`
- Body: The actual webpage code.

4. Visual Diagrams

Diagram 1: Standard Exam Diagram (Non-Persistent vs Persistent HTTP Connection)

Diagram 2: Easy Understanding Diagram (Bookstore Phone Order Analogy)

5. Comparison: HTTP vs HTTPS

Feature	HTTP (Hypertext Transfer Protocol)	HTTPS (HTTP Secure)

Security	Plain text (anyone on Wi-Fi can spy).	Encrypted using SSL/TLS (100% secure).
Port Number	Default Port **80**.	Default Port **443**.
Digital Certificate	No certificate needed.	Requires an SSL/TLS Certificate signed by a CA.
Performance	Slightly faster (zero encryption math).	Very fast on modern CPUs with TLS 1.3.
Browser Indicator	Marked as "Not Secure".	Displays padlock symbol in address bar.

5. Visual Examination & Concept Diagrams

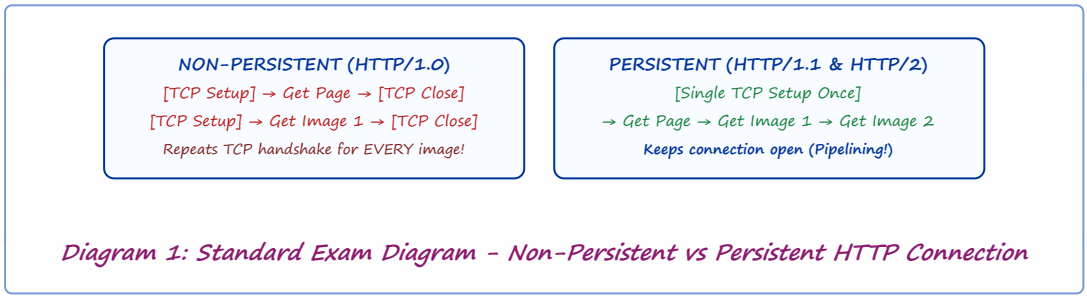


Diagram 1: Standard Exam Diagram - Non-Persistent vs Persistent HTTP Connection

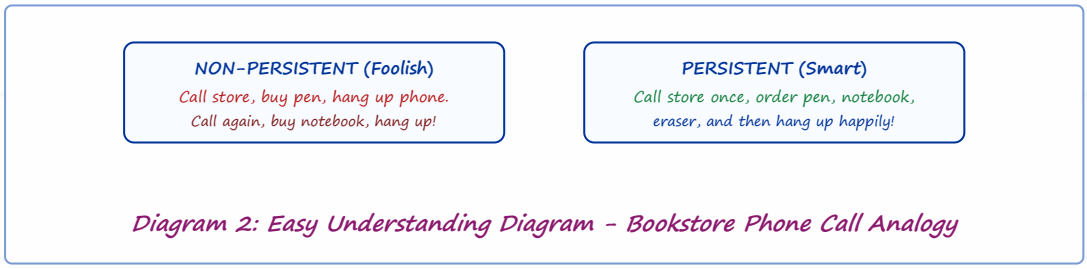


Diagram 2: Easy Understanding Diagram - Bookstore Phone Call Analogy

6. Summary Table

HTTP Version	Year	Major Improvement
HTTP/1.0	1996	Non-persistent connections (1 TCP connection per file).
HTTP/1.1	1997	Persistent connections and pipelining (Single connection for all assets).
HTTP/2	2015	Binary framing, multiplexing, and header compression over one stream.
HTTP/3	2022	Built on UDP using QUIC to eliminate head-of-line blocking.

7. Conclusion

HTTP is the universal communication protocol of the Web. Combined with TLS encryption (HTTPS) and persistent multiplexing, it enables fast, secure, and interactive web browsing for

the entire planet.
